AI-Assisted Interior Furniture Layout System - Course Project

Instructor: Prof. Ali Moridnejad Team Members: Qiyun Ge (Grant Ge), Yuping Yan
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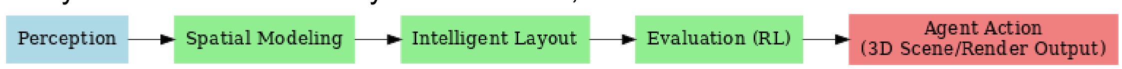
1. Motivation (Why it Matters)

- Professional: Automates layout iteration, saving designers time.
- Everyday: Helps families preview furniture fit and style, reducing waste.
- Academic: Combines AI, optimization, and HCI for real-world design.

2. System Architecture (Three Layers, Five Modules)

The system follows a three-layer architecture, subdivided into five modules.



3. Scope

In Scope (Core):

- 2D floor plan input
- Basic furniture placement (drag/drop, no overlap)
- Simple evaluation (space utilization, corridor width)
- Simple interface for visualization

Optional Extensions (if time allows):

- Simplified 3D visualization
- Automated heuristic-based layout generation
- Extended evaluation metrics

4. Timeline (Aligned with Course Schedule)

Dates	Activities	Deliverables
Sep 29 – Oct 3	Finalize proposal	Project Proposal
Oct 6 – Oct 24	Prototype Phase I	Early prototype demo;
Oct 27 – Oct 31	Refine prototype	Progress Report
Nov 3 – Nov 21	Prototype Phase II	Extended prototype features
Nov 24 – Nov 28	Refinement	Refined prototype
Dec 8	Present and submit final work	Final Report & Presentation

5. Evaluation Criteria (How Success Will Be Measured)

- Functionality: Does the prototype allow furniture placement without overlaps and display layouts?
- Usability: Is the interface simple and intuitive for non-experts?
- Robustness: Can it handle different room sizes and multiple furniture items?